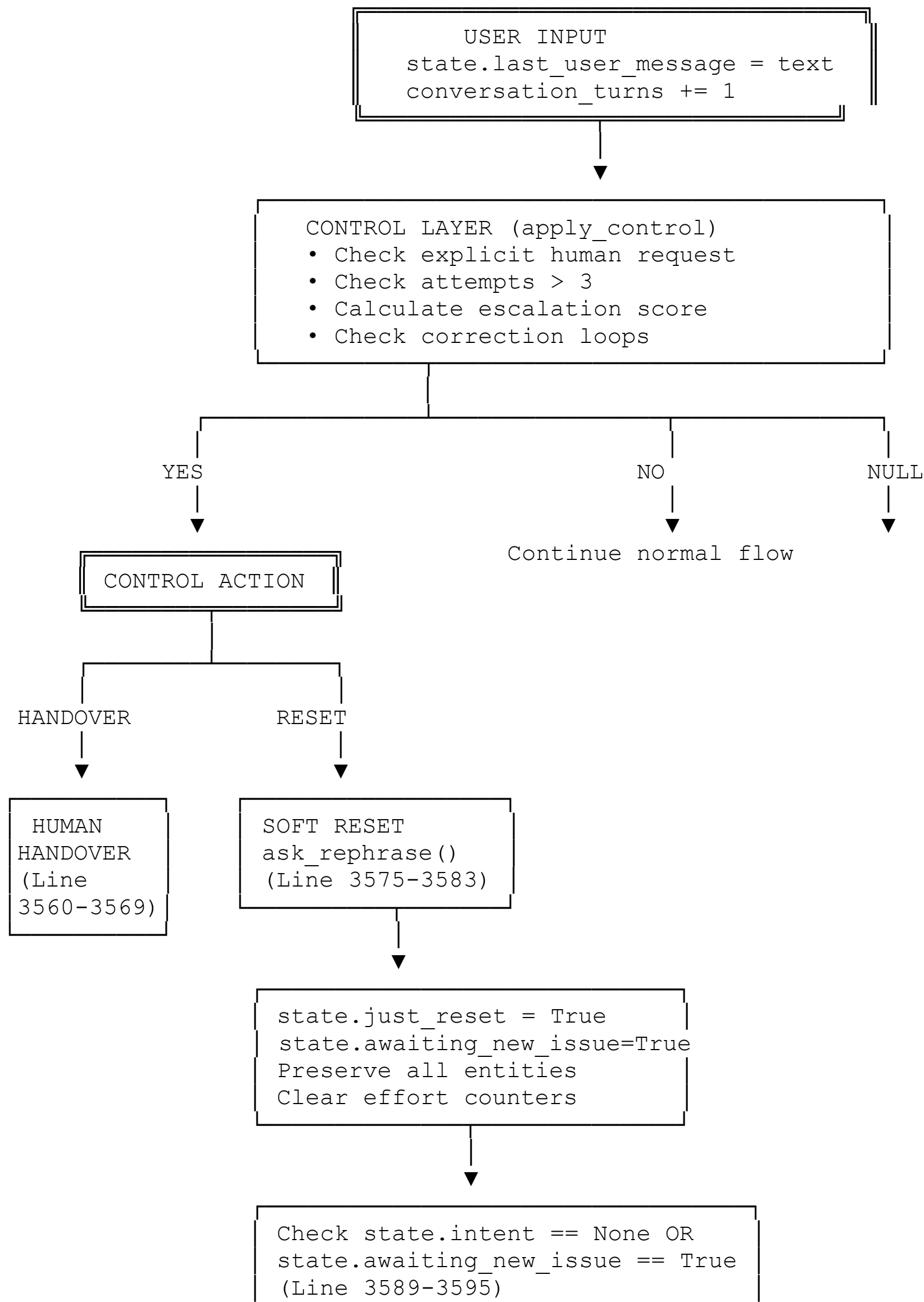
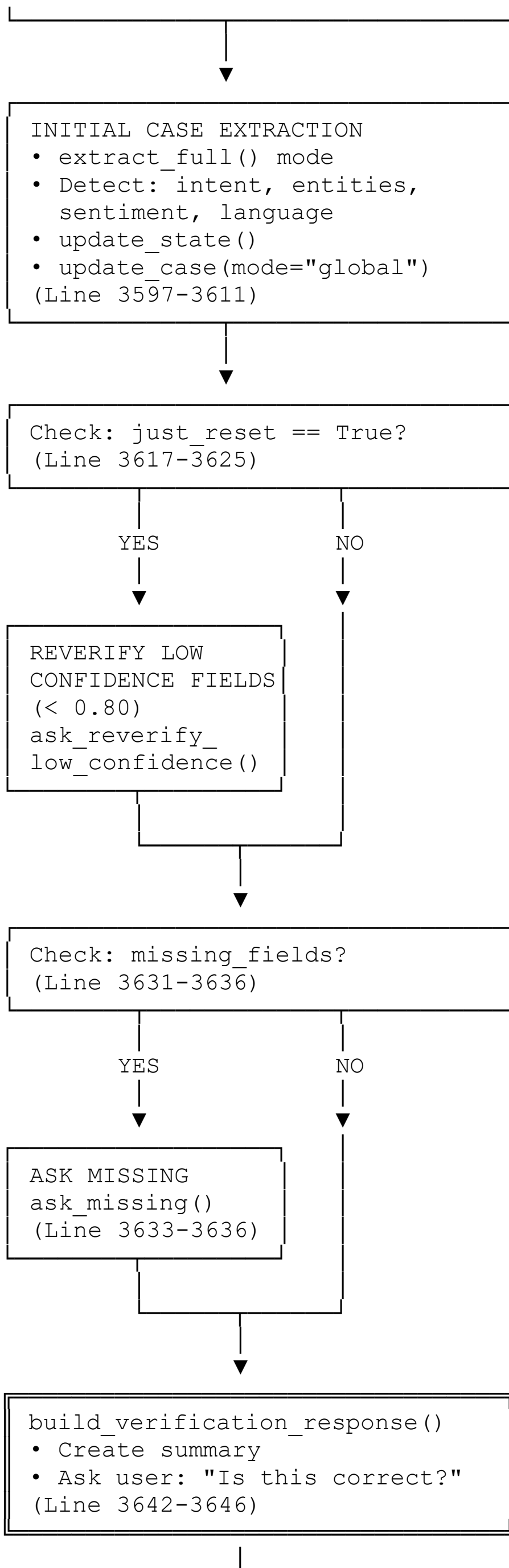


CLEAN AGENT FLOW DIAGRAMS WITH CODE EXPLANATIONS

DIAGRAM 1: MAIN AGENT PIPELINE FLOW





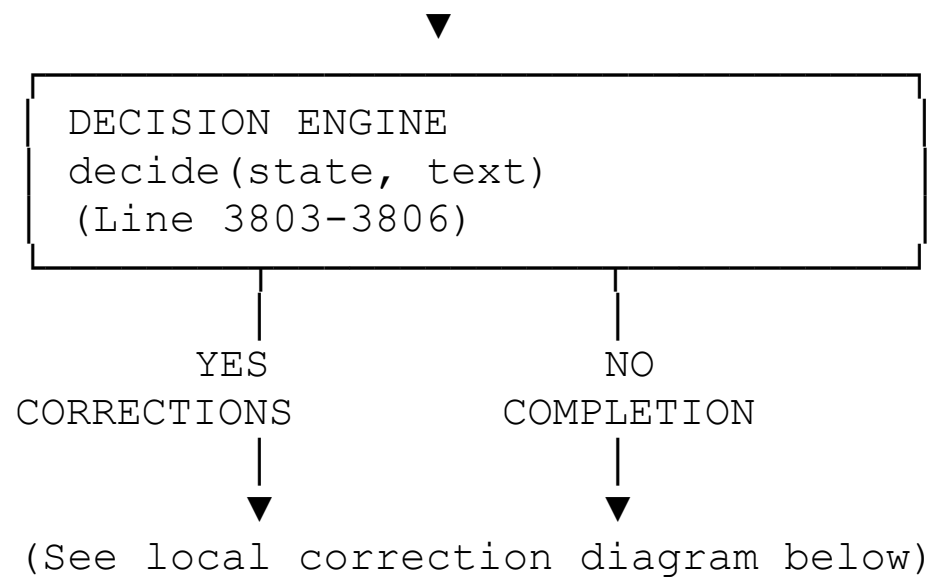
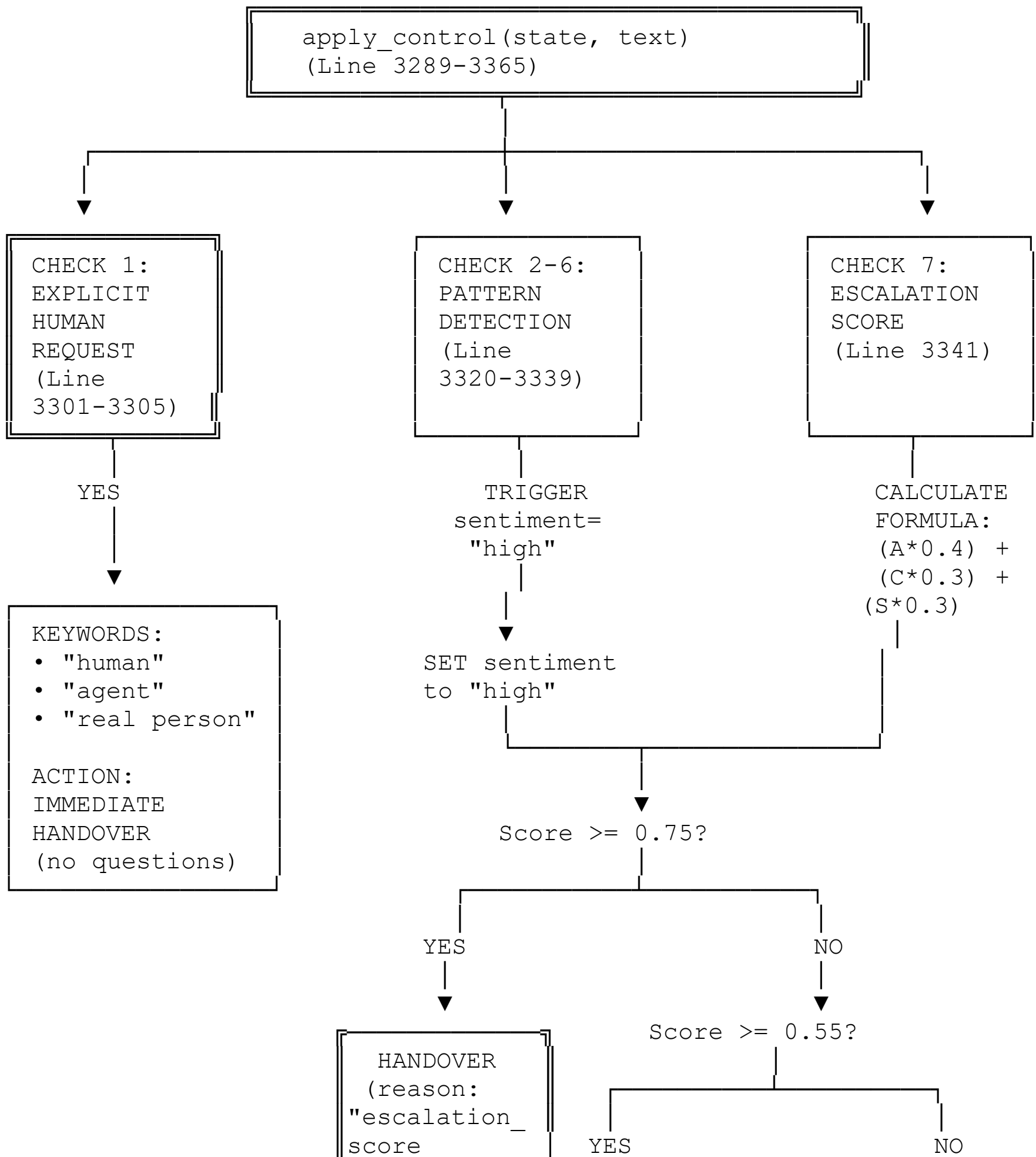
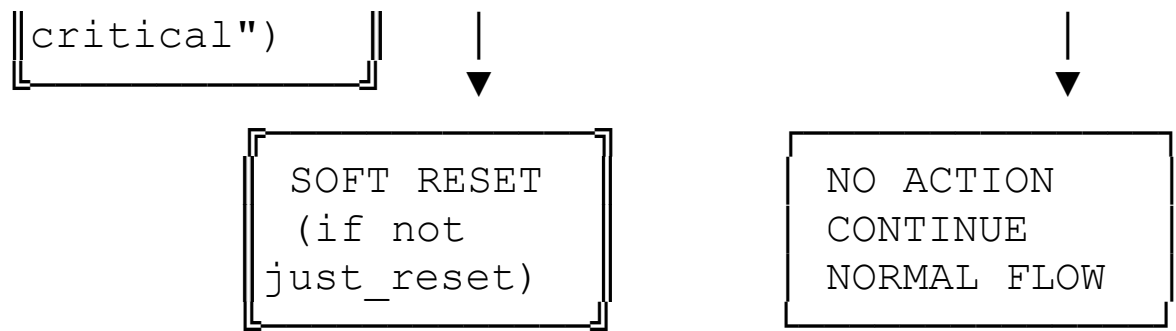


DIAGRAM 2: CONTROL LAYER - ESCALATION DECISION





ADDITIONAL CHECK: CORRECTION LOOP (Line 3355-3363)

```
if (awaiting_correction AND attempts >= 2 AND NOT just_reset):
```

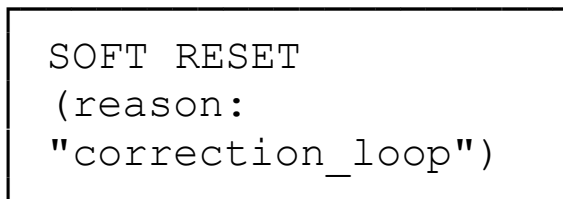
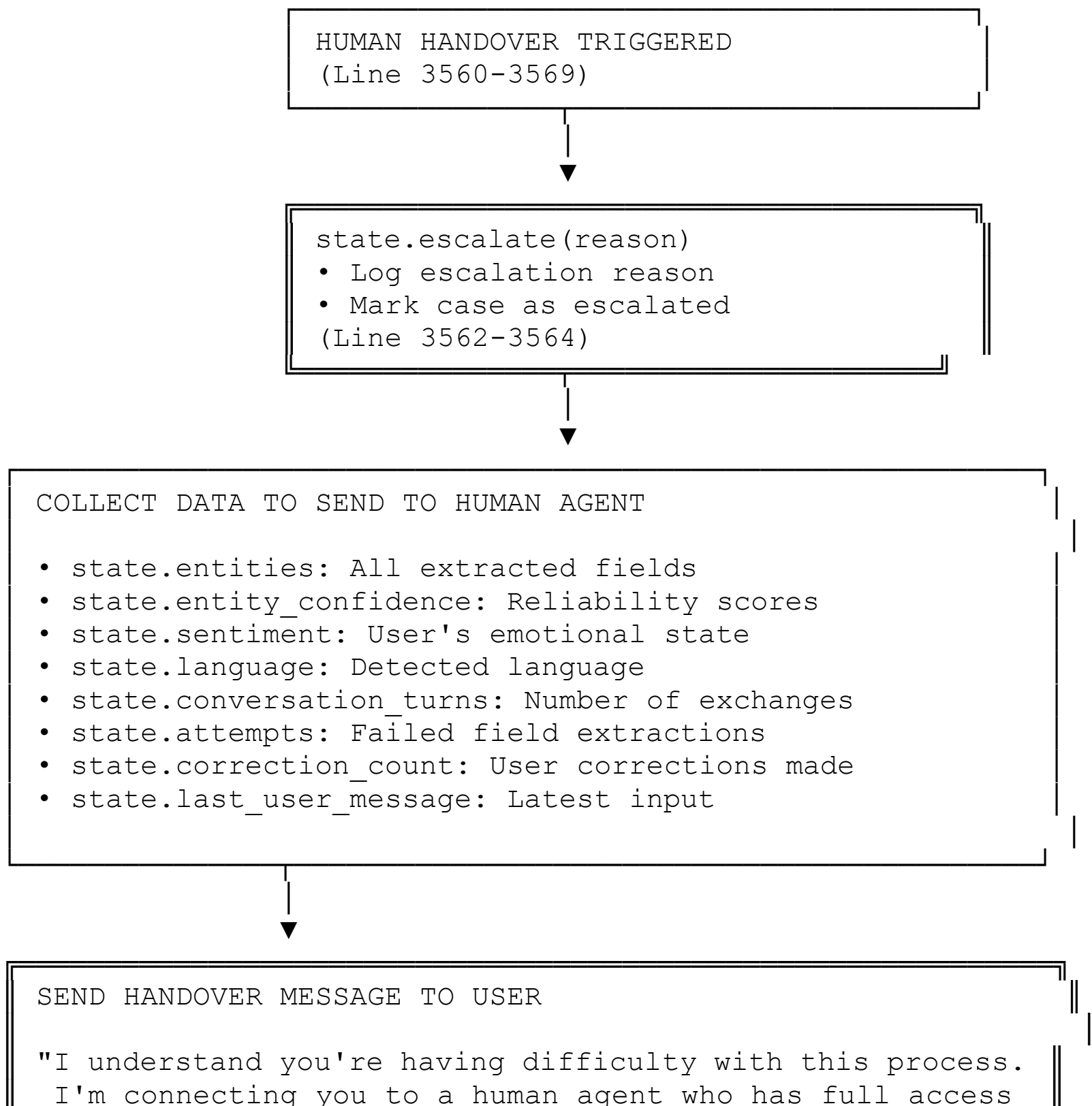


DIAGRAM 3: HUMAN HANDOVER FLOW



```
to your account and can provide personalized assistance. ||
Please hold while I transfer you... ||
action: "handover" ||
```



```
AGENT PIPELINE STOPS
• No more automated responses
• Human agent takes over
• Case marked as: ESCALATED / IN_PROGRESS
```



```
HUMAN AGENT RECEIVES:

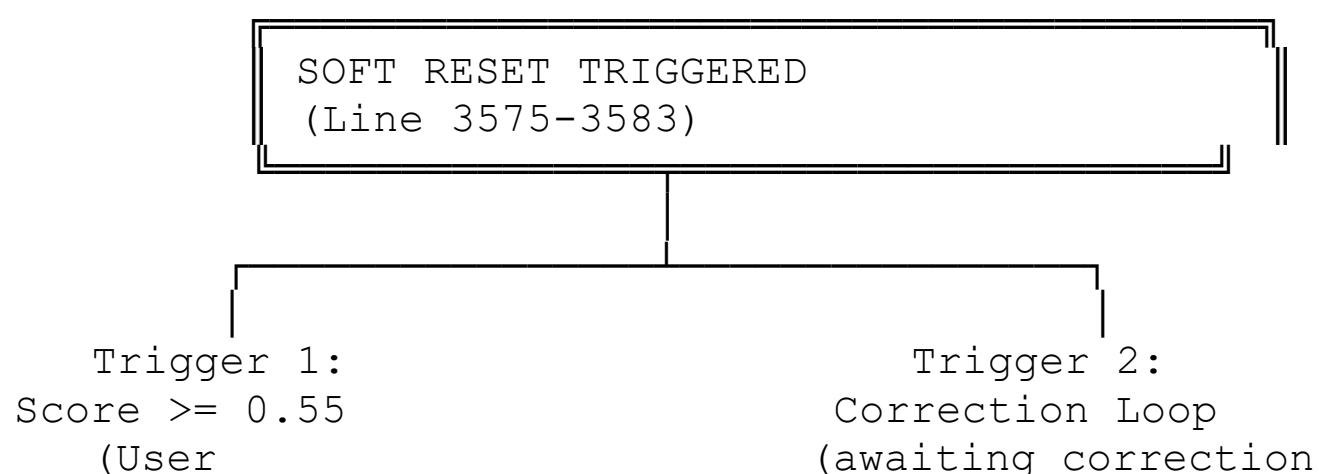
✓ Full conversation history
✓ All extracted data (name, email, issue, etc.)
✓ Confidence scores for each field
✓ Sentiment analysis (high frustration?)
✓ Number of attempts made by user
✓ Why escalation was triggered
✓ User's preferred language

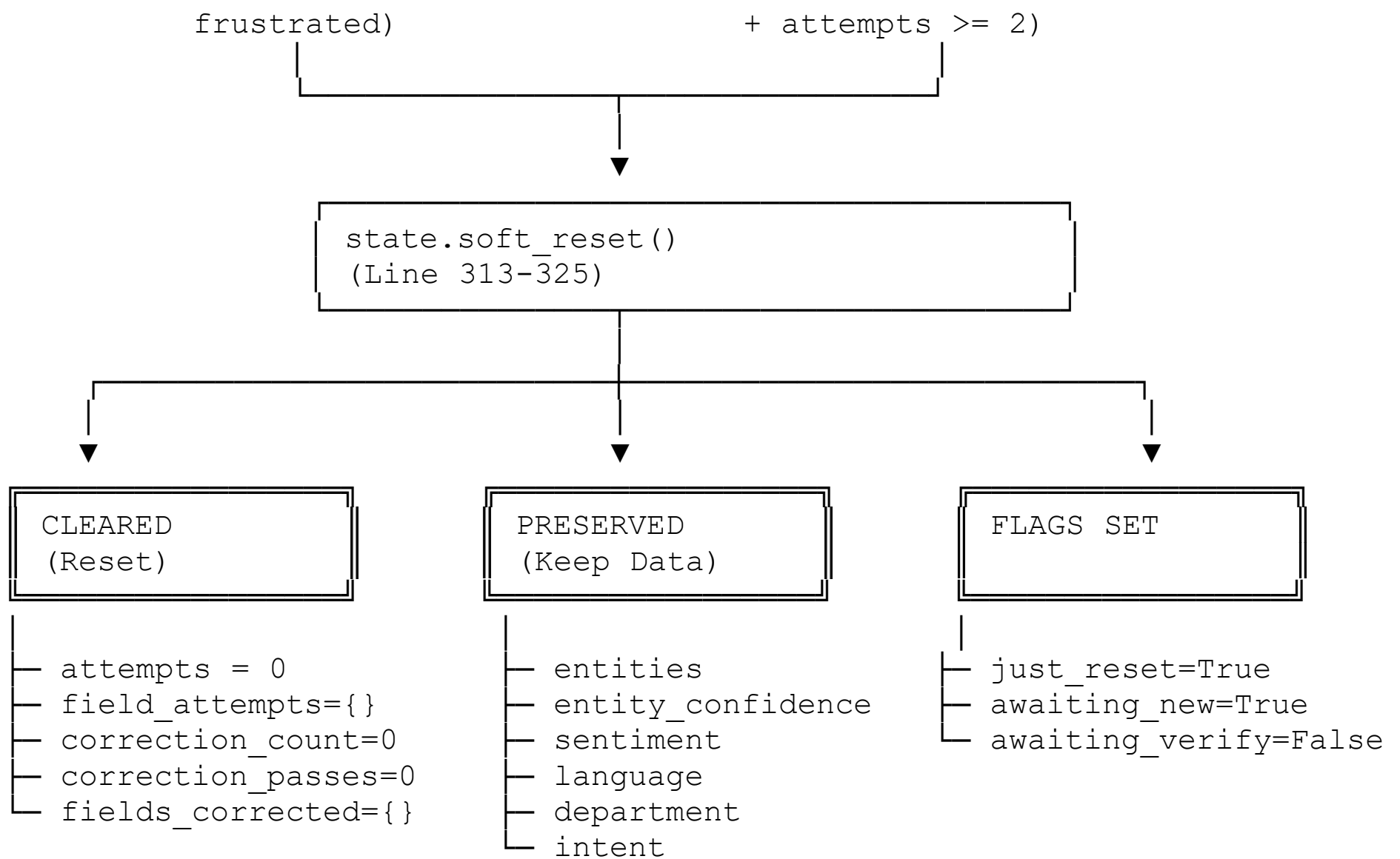
Human can then:
• Verify information directly with user
• Make final decision on case
• Process complaint immediately
• Provide personalized resolution
• Update case status
```

TRIGGERS FOR HANDOVER (All lead here):

1. Explicit request: "I want to talk to a human"
2. Attempts exceeded: attempts > 3
3. Escalation critical: score >= 0.75
4. Risk keywords: "assault", "abuse", "threat", etc.

DIAGRAM 4: SOFT RESET FLOW





EFFECT: User gets fresh start with cleared effort,
but system remembers extracted data & context



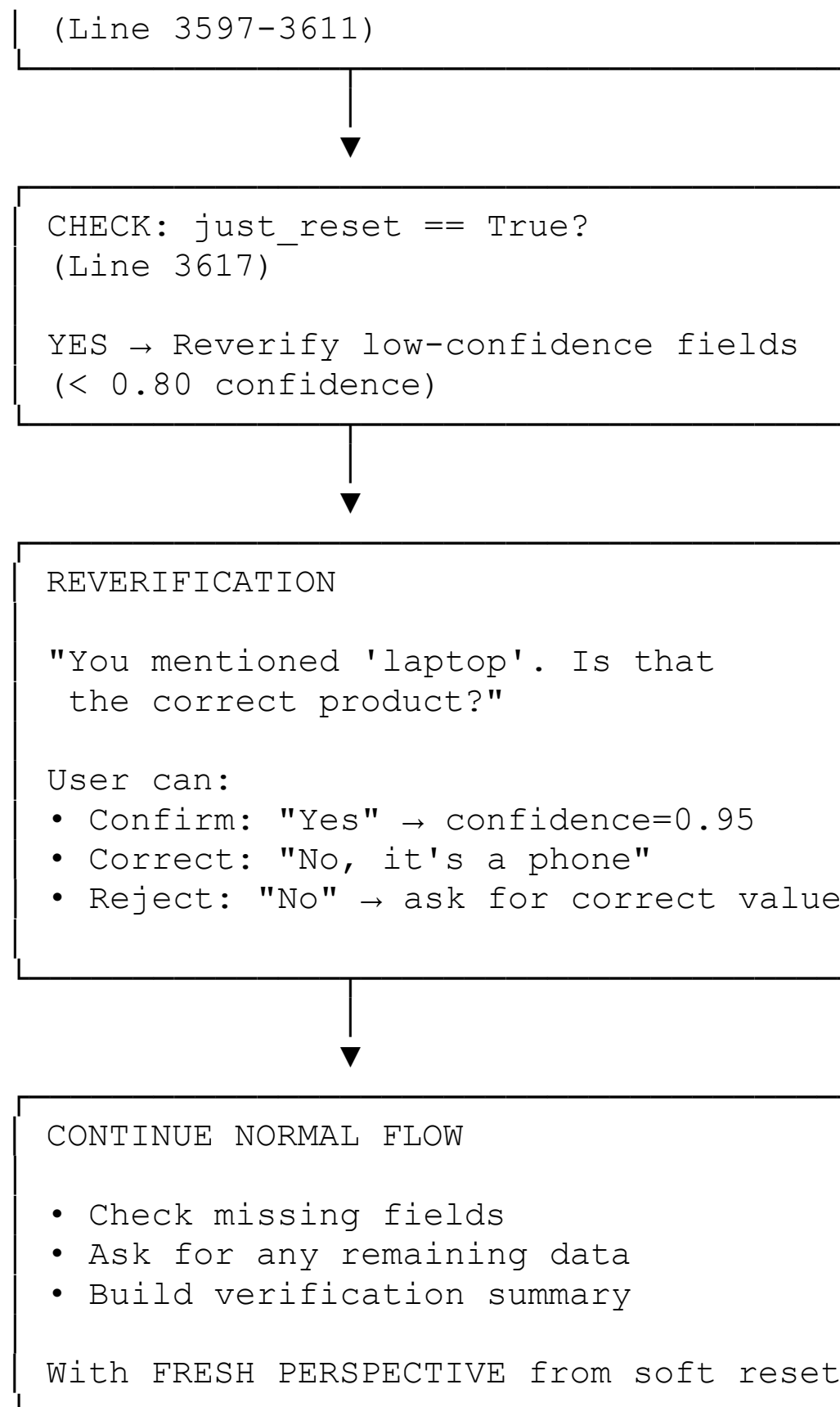
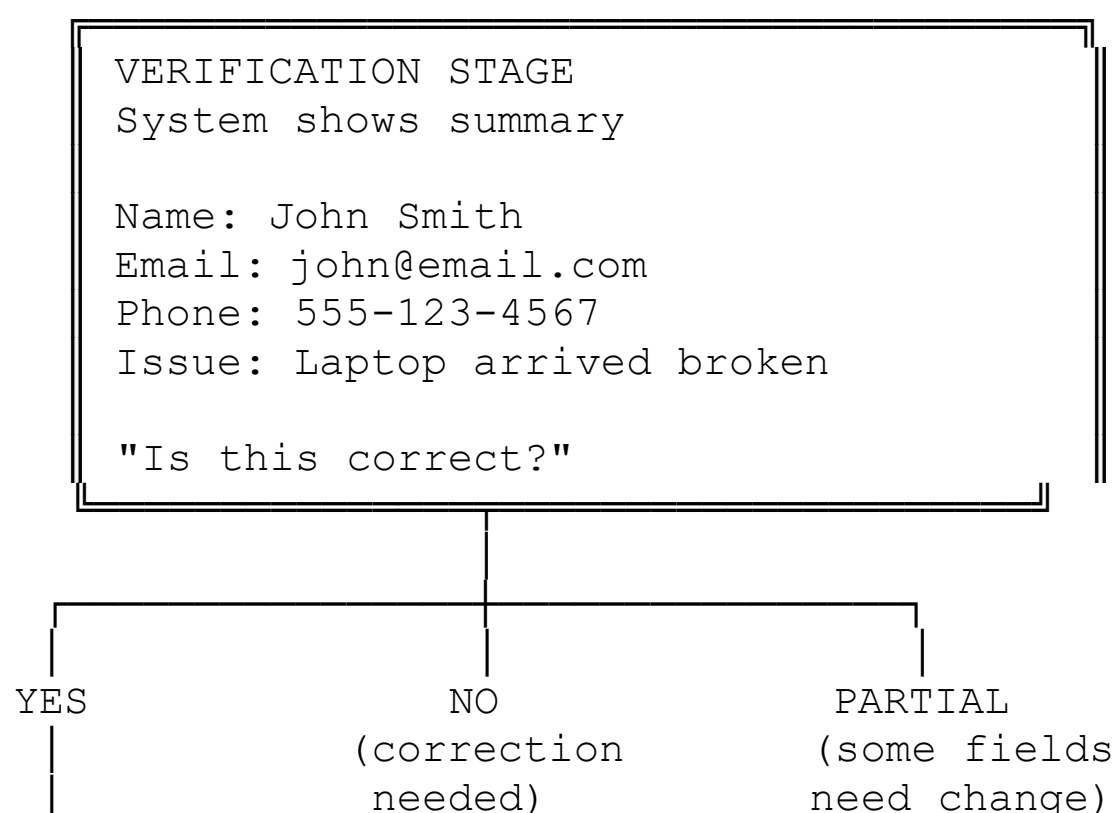
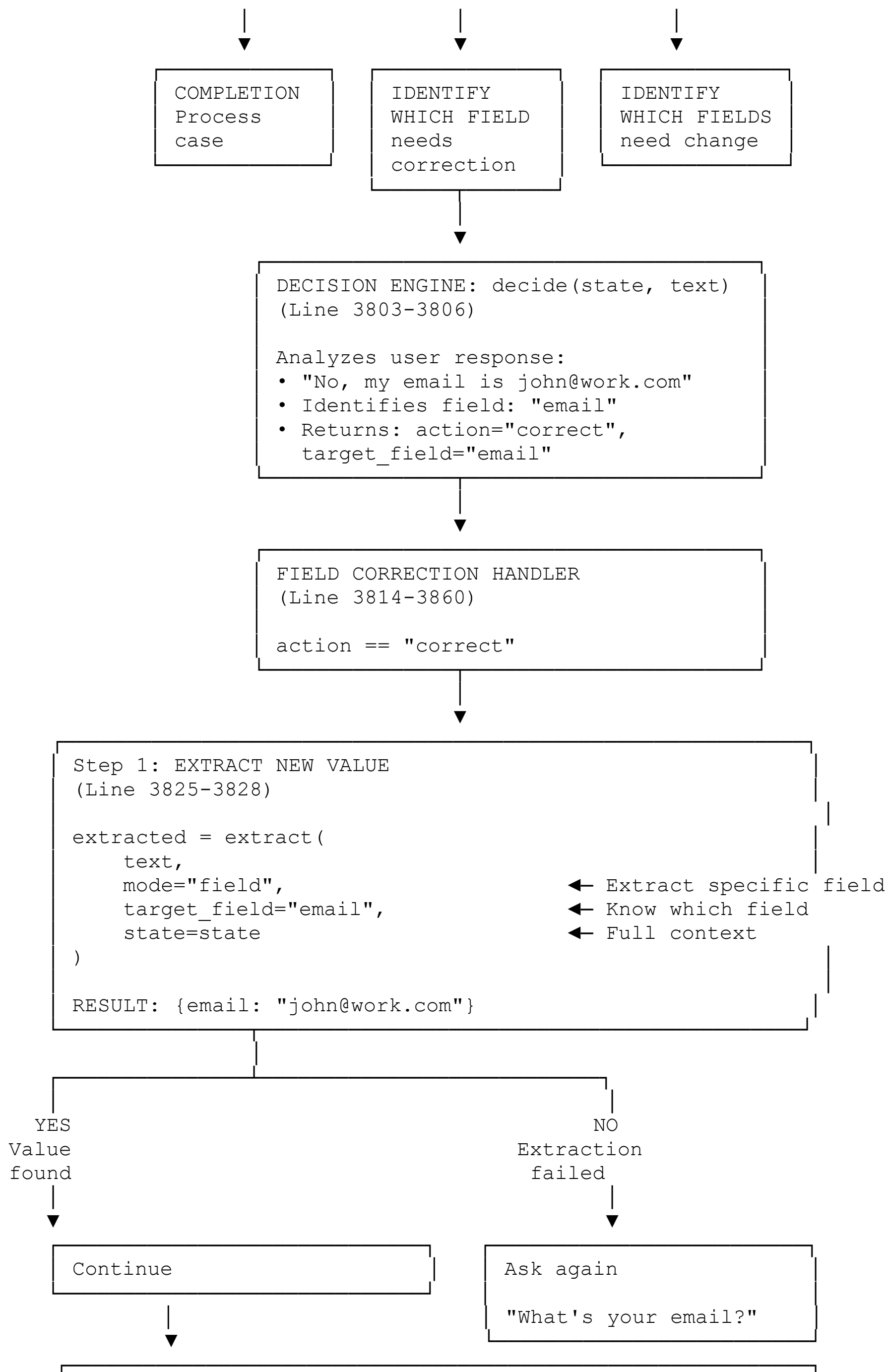


DIAGRAM 5: LOCAL CORRECTION FLOW (During Verification)





Step 2: UPDATE CASE STATE
(Line 3833-3837)

```
state = update_case(  
    state,  
    extracted,  
    mode="normal"           ← Normal correction mode  
)
```

This will:

- Update entity: `entities["email"] = "john@work.com"`
- Increment `correction_count += 1`
- Set confidence for email field
- Check for correction drift



Step 3: BOOST CONFIDENCE
(Line 3847)

```
state.increase_confidence(0.02)  
• Confidence += 0.02  
• Shows system trusts the correction  
• Caps at 1.0 (100%)
```



Step 4: RESET ATTEMPT COUNTERS
(Line 3849-3852)

```
state.reset_attempts()  
• Clear attempts counter for this field  
• Allows clean re-extraction if needed  
• Prevents artificial "stuck" state
```



Step 5: CHECK FOR MORE CORRECTIONS
(Line 3859-3860)

```
if state.missing_fields:  
    return ask_missing()  
else:  
    return build_verification_response() ← Show summary
```

OPTIONS:

- More fields to collect → `ask_missing()`
- All done → build summary (goto verification)



Step 6: CHECK CORRECTION DRIFT
(Automatic in `apply_control`)

During this flow, `apply_control()` checks:

- Same field corrected 3+ times?
→ sentiment = "high"
- Corrections > field count?
→ sentiment = "high"
- Multi-pass drift?
→ sentiment = "high"

If triggered:

- `escalation_score` >= 0.55
- SOFT RESET triggered again



SHOW UPDATED SUMMARY (if no drift)

Name: John Smith

Email: john@work.com ← CORRECTED

Phone: 555-123-4567

Issue: Laptop arrived broken

"Is this correct now?"

User can:

- "Yes" → COMPLETION
- "No, X is still wrong" → Another correction